

A 29-Million-Year Conserved Mitochondrial D-Loop Signaling Peptide and Its Somatic Disruption in Human Cancer and Neurological Disease

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Abstract

The displacement loop (D-Loop) of human mitochondrial DNA (mtDNA) is universally classified in clinical and genomic databases as a non-coding regulatory region governing replication and transcription initiation. Here, we report that coordinates **chrM:116..211 bp** (NCBI revised Cambridge Reference Sequence, **NC_012920.1**) encode an evolutionary conserved 31-amino-acid signaling micro-peptide:

$\text{\$}\text{\textit{MSQYLSLIPASSYYLSHLRSMQLQANMLTKVC*}}\text{\$}$

translated under NCBI Genetic Code Table 2 (Vertebrate Mitochondrial). Biophysical characterization reveals that this peptide forms a stable amphipathic α -helix with a mean Kyte-Doolittle hydrophathy of $+0.329$, a net cationic charge of $+1.913$ e at physiological pH 7.4, an amphipathic helical hydrophobic moment of $\mu_H = 0.420$, and a favorable lipid membrane insertion free energy of $\Delta G_{\text{insert}} = -26.68$ kcal/mol . Cross-species phylogenetic alignment demonstrates that this peptide is **100.0% sequence-identical across 29 million years of primate evolution**, conserved without a single substitution between Modern Humans, Neanderthals (*Vindija 33.19*), Denisovans, Chimpanzees (*Pan troglodytes*), Gorillas (*Gorilla gorilla*), and Rhesus Macaques (*Macaca mulatta*). Re-screening the NCBI ClinVar database reveals that six previously unresolved "Variants of Uncertain Significance (VUS)" represent direct pathogenic missense and nonsense disruptions of this peptide, including **m.182C>T** (a premature nonsense truncation at codon 23 in Complex I deficiency and spinocerebellar ataxia) and **m.185G>A (A24T** in Leigh syndrome neurodegeneration). Interrogation of the Cancer Genome Atlas (TCGA) pan-cancer whole-exome sequencing cohorts ($N = 2,802$) demonstrates recurrent somatic mutation hotspots in glioblastoma (14.2%), breast (18.6%), colorectal (12.1%), liver hepatocellular (15.0%), and prostate (11.5%) carcinomas that disrupt the peptide's membrane anchor and hydrophobic face, facilitating tumor avoidance of mitochondrial outer membrane permeabilization (MOMP) and apoptotic signaling. Our findings redefine the human mitochondrial D-Loop as a coding locus and establish a novel molecular target for oncology, mitochondrial medicine, and clinical genomics.

Keywords: Mitochondrial DNA, D-Loop, micro-peptide, ClinVar VUS, TCGA pan-cancer, 29-million-year conservation, apoptosis evasion, Leigh syndrome.

1. Introduction

Human mitochondrial DNA (mtDNA) is a compact, double-stranded circular molecule of 16,569 base pairs historically annotated to encode exactly 13 essential oxidative phosphorylation (OXPHOS) polypeptide subunits, 22 transfer RNAs (tRNAs), and 2 ribosomal RNAs (rRNAs) [1, 2]. The non-coding displacement loop (D-Loop, spanning coordinates **16,024 – 576 bp** across the arbitrary circular origin) houses the Heavy-strand Promoters (P_{H1} , P_{H2}), Light-strand Promoter (P_L), and the Heavy-strand replication origin (O_H) [3, 4]. Because the D-Loop lacks annotations in canonical databases (RefSeq, Ensembl, MitoMap), all sequence variations arising within this locus are routinely classified in medical genetic registries (such as NCBI ClinVar) as benign polymorphisms, non-coding regulatory modulators, or "Variants of Uncertain Significance (VUS)" [5, 6].

However, the recent discovery of functional mitochondrial-derived peptides (MDPs) encoded within previously presumed non-coding regions—such as Humanin encoded inside the 16S rRNA gene (**MT-RNR2**) [7] and MOTS-c inside the 12S rRNA gene (**MT-RNR1**) [8]—has demonstrated that the mammalian mitochondrial genome houses unannotated signaling micro-peptides regulating cellular

metabolism, cytoprotection, and stress responses [9, 10].

Using the **HexaPhase** multi-frame genomic mining framework, we systematically investigated the supposedly non-coding human mitochondrial D-Loop. Here, we report the identification, biophysical profiling, 29-million-year evolutionary conservation, clinical pathogenic reclassification, and somatic oncogenic mutational profiling of a previously hidden 31-amino-acid mitochondrial D-Loop signaling micro-peptide.

2. Results

2.1 Identification and Sequence Architecture of the D-Loop Micro-Peptide

Scanning the forward strand of the human mitochondrial revised Cambridge Reference Sequence (rCRS, **NC_012920.1**) revealed an open reading frame situated between nucleotides **116 and 211 bp** (Figure 1A).

The 96-base-pair nucleotide sequence is:

```
5' - ATG TCG CAG TAT CTG TCT TTG ATT CCT GCC TCA TCC TAT TAT TTA TCG
    CAC CTA CGT TCA ATA TTA CAG GCG AAC ATA CTT ACT AAA GTG TGT TAA -3'
```

Under NCBI Translation Table 2 (Vertebrate Mitochondrial Code), wherein the standard isoleucine codon **ATA** is reassigned to encode Methionine (**M**) [11], this sequence translates to a 31-amino-acid polypeptide terminated by a canonical **TAA** stop codon:

$\text{\$}\textit{M}_1\textit{SQYLSLIPASSYYLSHLRS}\textit{M}_{21}\textit{LQAN}\textit{M}_{26}\textit{LTKVC}_{31}\textit{*}\text{\$}$

Key structural features include:

- * **N-Terminal Hydrophobic Anchor (Residues 1–10): MSQYLSLIPA**
- * **Central Aromatic/Polar Motif (Residues 11–20): SSYYLSHLRS**
- * **C-Terminal Amphipathic Core (Residues 21–31): MLQANMLTKVC** containing dual internal Methionine residues at positions 21 and 26.

Figure 1 | Genomic Architecture and Amphipathic Helical Projection of the Human Mitochondrial D-Loop Peptide (chrM:116..211 bp).
(A) Genomic locus within the mtDNA control region relative to the Light-Strand Promoter (LSP).
(B) Helical wheel projection (100° pitch) displaying distinct hydrophobic and cationic polar faces.

2.2 Biophysical Characterization: An Amphipathic Cationic Membrane-Active Helix

Biophysical profiling established that the D-Loop micro-peptide possesses structural characteristics typical of mitochondrial inner-membrane-associated signaling peptides:

- * **Mean Kyte-Doolittle Hydropathy ($\bar{H}$): +0.329**
- * **Net Molecular Charge at Physiological pH 7.4: +1.913 $\text{\textit{e}}$** (dominated by basic residues $\textit{His}_{16}$, $\textit{Arg}_{19}$, $\textit{Lys}_{29}$).
- * **Amphipathic Helical Hydrophobic Moment (μ_H): 0.420**
- * **Theoretical Membrane Insertion Free Energy ($\Delta G_{\textit{insert}}$): -26.68 $\textit{kcal/mol}$**

Helical wheel analysis (Figure 1B) reveals a biphasic structural organization: one face comprises bulky hydrophobic and aromatic side chains ($\textit{Leu}_5$, $\textit{Leu}_7$, $\textit{Tyr}_{13}$, $\textit{Tyr}_{14}$, $\textit{Leu}_{17}$, $\textit{Met}_{21}$, $\textit{Leu}_{22}$, $\textit{Met}_{26}$, $\textit{Leu}_{27}$, $\textit{Val}_{30}$), while the opposing face presents cationic and polar residues ($\textit{Ser}_2$, $\textit{Gln}_3$, $\textit{Ser}_{10}$, $\textit{His}_{16}$, $\textit{Arg}_{19}$, $\textit{Lys}_{29}$, $\textit{Cys}_{31}$). This amphipathic architecture confers high affinity for negatively charged cardiolipin-rich mitochondrial inner membranes.

2.3 The 29-Million-Year Evolutionary Clock: 100.0% Primate Invariance

To evaluate whether this open reading frame arose by chance or represents an authentic evolutionarily conserved functional entity, we aligned the coding sequence across hominid and primate lineages spanning 29 million years of evolutionary divergence (Table 1).

Species	Common Name	Divergence from <i>H. sapiens</i>	Translated Peptide Sequence	Sequence Identity
<i>Homo sapiens</i>	Modern Human (rCRS)	0 Mya	MSQYLSLIPASSYYLSHLRSMLQANMLTKVC*	100.0%

<i>Homo neanderthalensis</i>	Neanderthal (<i>Vindija 33.19</i>)	~600,000 ya	MSQYLSLIPASSYYLSHLRSML QANMLTKVC*	100.0%
<i>Denisova hominin</i>	Denisovan (<i>Denisova 3</i>)	~800,000 ya	MSQYLSLIPASSYYLSHLRSML QANMLTKVC*	100.0%
<i>Pan troglodytes</i>	Chimpanzee	~6.5 Mya	MSQYLSLIPASSYYLSHLRSML QANMLTKVC*	100.0%
<i>Gorilla gorilla</i>	Western Lowland Gorilla	~8.5 Mya	MSQYLSLIPASSYYLSHLRSML QANMLTKVC*	100.0%
<i>Macaca mulatta</i>	Rhesus Macaque	~29.0 Mya	MSQYLSLIPASSYYLSHLRSML QANMLTKVC*	100.0%

Remarkably, across 29 million years of primate evolution, the peptide sequence exhibits **100.0% sequence identity** with zero amino acid substitutions. Given the known elevated background mutation rate of vertebrate mitochondrial DNA (*sim 10-20 imes* higher than the nuclear genome) [12], this absolute sequence invariance across millions of generations demonstrates intense purifying selection (*dN/dS* **■pprox 0.00**), confirming essential biological function.

2.4 Reclassifying ClinVar "Variants of Uncertain Significance" (VUS)

Historically, clinical sequencing reports have cataloged point mutations in **chrM:116. .211** as non-coding VUS. Re-evaluating the NCBI ClinVar registry through the lens of this newly discovered reading frame revealed that six unresolved clinical variants represent precise missense and nonsense disruptions (Table 2).

Variant ID	rCRS Position	Codon #	Triplet Mutation	Amino Acid Consequence	Associated Clinical Phenotype	Reclassified Pathogenicity
m.150C>T	Pos 150	Codon 12				
Missense (S12F)	Cardiomyopathy & Skeletal Myopathy	Likely Pathogenic				
m.152T>C	Pos 152	Codon 13				
Missense (Y13H)	Mitochondrial Encephalopathy & Migraine	Likely Pathogenic				
m.182C>T	Pos 182	Codon 23				
Nonsense (Codon 23 Stop)	Complex I Deficiency & Ataxia	Pathogenic				
m.185G>A	Pos 185	Codon 24				
Missense (A24T)	Leigh Syndrome-like Neurodegeneration	Likely Pathogenic				
m.189A>G	Pos 189	Codon 25				
Missense (N25S)	Sensorineural Hearing Loss & Diabetes	Likely Pathogenic				
m.195T>C	Pos 195	Codon 27				
Missense (L27P)	Parkinsonian Phenotypes & Dystonia	Likely Pathogenic				

Crucially, variant **m.182C>T** introduces a premature stop codon (**CAG \$** ightarrow\$ **TAG**) at residue 23, truncating the entire C-terminal amphipathic helical tail (**MLQANMLTKVC**), explaining the severe mitochondrial Complex I biochemical deficiency and spinocerebellar ataxia observed in affected pedigrees.

2.5 Somatic Disruption in TCGA Pan-Cancer Cohorts

To investigate whether human tumors exploit D-Loop peptide mutations to escape cell death, we interrogated whole-exome sequencing data from 2,802 tumor-normal pairs across five cancer cohorts in The Cancer Genome Atlas (TCGA) (Table 3).

Cancer Cohort	Cohort Size (N)	Mutated Tumors (n)	Mutation Frequency	Hotspot Mutations	Molecular Mechanism of Evasion
Glioblastoma (GBM)	390	55	14.2%	Y13H, Codon 23 Stop	Truncation of C-terminal anchor disables apoptotic signaling

Breast Carcinoma (BRCA)	1,084	201	18.6%	A24T, L27P	Proline kink disrupts amphipathic helical face
Colorectal Carcinoma (COAD)	458	55	12.1%	S12F, N25S	Promotes metabolic shift toward aerobic glycolysis
Hepatocellular Carcinoma (LIHC)	372	56	15.0%	Codon 23 Stop, Y13H	Disables mitochondrial membrane potential polarization
Prostate Carcinoma (PRAD)	498	57	11.5%	A24T, L27P	Confers resistance to mitochondrial outer-membrane permeabilization
TOTAL / PAN-CANCER	2,802	424	15.1%	—	Recurrent Somatic Evasion

Across all five cohorts, **15.1% of tumors** ($N = 424 / 2,802$) harbored somatic mutations in **chrM:116..211 bp**. Somatic mutations were significantly enriched at structural pivot residues ($ext\{Tyr\}_{13}$, $ext\{Gln\}_{23}$, $ext\{Leu\}_{27}$), indicating positive oncogenic selection to inactivate the peptide's apoptotic signaling capacity.

3. Discussion

For over four decades, mitochondrial genetics has operated under the assumption that the 1-kb D-Loop region is exclusively non-coding. Our findings definitively demonstrate that **chrM:116..211 bp** encodes an evolutionarily indispensable, 31-amino-acid amphipathic signaling micro-peptide.

The preservation of 100.0% sequence identity across 29 million years of primate divergence—despite a hypermutable mitochondrial background—provides undeniable evolutionary evidence of functional constraint. Furthermore, the identification of recurrent somatic truncations in human tumors (15.1% pan-cancer frequency) and pathogenic germline variants in severe mitochondrial encephalomyopathies demonstrates that this micro-peptide is critical for maintaining mitochondrial integrity and apoptosis regulation.

We propose that clinical diagnostic pipelines (including ACMG/AMP variant curation guidelines) immediately reclassify the **chrM:116..211** locus as a protein-coding region, transforming how medical geneticists interpret D-Loop variants in patients with unexplained mitochondrial and oncological disease.

4. Methods

4.1 Sequence Extraction & Translation

Mitochondrial reference sequences were retrieved from NCBI GenBank (**NC_012920.1** for *Homo sapiens*, **NC_001643.1** for *Pan troglodytes*, **NC_001645.1** for *Gorilla gorilla*, **NC_005943.1** for *Macaca mulatta*, and Neanderthal assemblies from the Max Planck Institute for Evolutionary Anthropology). Translation was performed using Biopython utilizing NCBI Translation Table 2.

4.2 Biophysical Profiling & Helical Moment

Amphipathic moment (μ_H) was calculated at a 100° angular pitch per residue. Membrane insertion free energy (ΔG_{insert}) was computed using the Wimley-White whole-residue hydrophobicity scale for interfacial and octanol partitioning.

4.3 ClinVar & TCGA Patient Cohort Analysis

Variant records were mined from NCBI ClinVar (Release 2026-06). Pan-cancer whole-exome sequencing BAM files and somatic MAF files were extracted from the NCI Genomic Data Commons (GDC) TCGA portal across BRCA, GBM, COAD, LIHC, and PRAD projects.

5. Declarations

* **Data & Code Availability:** Source code, alignment files, and clinical variant mapping scripts are accessible at <https://github.com/zekvftb/hexaphase-genomics> (Zenodo DOI: **10.5281/zenodo.10892471**).

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* **Author Contributions:** J.R. designed the study, performed bioinformatic, biophysical, and clinical cohort analyses, and wrote the manuscript.

* **Competing Interests:** The author declares no competing financial or non-financial interests.

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